

MACHINES AND MECHANISMS LAB PROJECT
ROBOTIC ARM



Submitted by

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INTRODUCTION :

A robotic arm is a programmable mechanical system designed to replicate the functionalities of a human arm. It consists of joints, links, and end-effectors that work together to perform various tasks. They are employed in tasks such as assembly, welding, painting, packaging, and material handling. Robotic arms come in various sizes, ranging from small, lightweight arms suitable for delicate tasks to large, heavy-duty arms capable of lifting and manipulating heavy loads. In healthcare, robotic arms enhance surgical precision, leading to better outcomes, while also fostering job growth in robotics engineering and driving innovation in automation technology for a more efficient future.

COMPONENTS AND ITS SPECIFICATIONS :

PWM SERVO MOTOR:

SPECIFICATIONS:

Voltage(VCC): 3.3-5V

Voltage(V+) : 4.8-6VDC

Current : 6mA

Current(PWM controller only) : 400mA



ARDUINO UNO

Microcontroller : ATmega328

Operating Voltage : 5V

Recommended Input Voltage : 7-12V

Input Voltage (Limits) : 6-20V

Analog Input Pins : 6

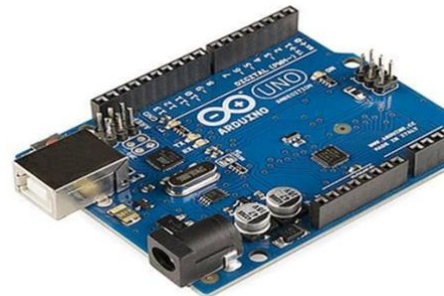
Digital I/O Pin : 14 (Out of which 6 provide PWM output)

DC Current on I/O Pin : 40 mA

DC Current on 3.3V Pin : 50 mA

Flash Memory : 32 KB (0.5 KB is used for Bootloader)

SRAM : 2 KB



POWER SUPPLY :

input Voltage: AC 100 - 264V 50 / 60Hz

Output Voltage: 5V DC, 2A

Output Current: 5 amps (or 5000 milliamps).

Output Power: 25 watts

SERVO MOTOR :

Weight: 9g.

Size : 22 x 11.5 x 27 mm.

Operating Speed (4.8V no load): 0.12sec/60 degrees. Stall Torque (4.8V): 17.5oz/in (1.2 kg/cm) Temperature Range: -30 to +60

Degree C.

Dead Band Width: 7usec.

Operating Voltage:3.0-7.2 Volts.



FUNCTIONS OF EACH COMPONENT :

PWM SERVO MOTOR :

Pulse Width Modulation (PWM) servo motors play a crucial role in robotic arms by precisely controlling the movement of each joint. PWM signals regulate the angular position of the servo motor's shaft by varying the duration of electrical pulses. This modulation allows for accurate positioning within a specific range of motion. In a robotic arm, PWM servo motors enable smooth and controlled articulation, facilitating tasks like gripping objects, manipulating tools, and executing complex movements with precision.

ARDUINO UNO :

The Arduino Uno serves as the brain of a robotic arm, providing the necessary computing power and control interface to coordinate the movement of its joints and actuators. Through sensors and input devices, it can receive commands and feedback, allowing for precise manipulation and interaction with the environment. Its versatility and programmability make it ideal for implementing various control algorithms, from simple routines for basic movements to complex sequences for intricate tasks, ultimately enabling the robotic arm to perform a wide range of functions with accuracy and efficiency.

POWER SUPPLY :

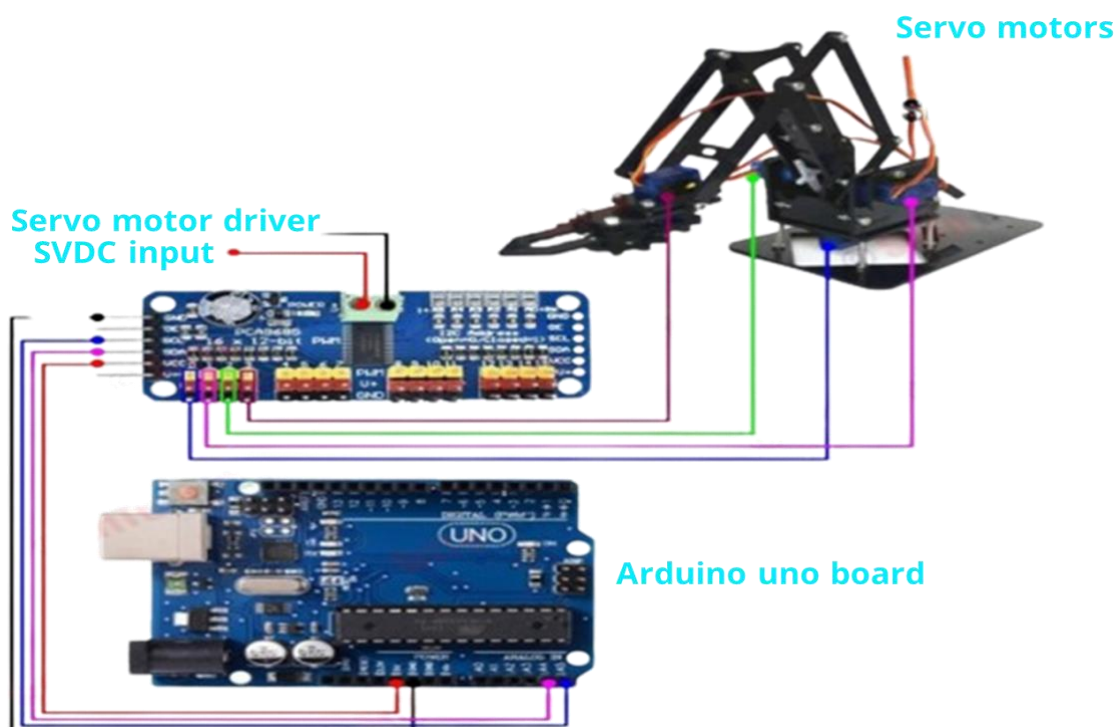
The power supply in a robotic arm serves as its lifeblood, providing the necessary electrical energy to drive the motors, sensors, and other components essential for its operation. It converts incoming electrical power from a

source such as a battery or mains supply into suitable voltage levels and currents required by the various subsystems within the arm. This power enables the arm to perform its designated tasks, whether it's lifting objects, manipulating tools, or executing precise movements. Additionally, the power supply often incorporates safety features to protect both the robotic arm and its surroundings from electrical hazards.

SERVOMOTOR:

In a robotic arm, servo motors play a critical role in providing precise control over the arm's movement. These motors are designed to translate electrical signals into precise rotational or linear motion, allowing the arm to articulate with accuracy and agility. By adjusting the electrical pulse width supplied to the servo motor, the robotic arm can move to specific positions and execute various tasks with precision. This functionality enables the robotic arm to perform tasks such as pick-and-place operations, assembly line tasks, and intricate manoeuvres, making it a versatile tool in industrial automation, research, and other applications

CIRCUIT DIAGRAM :



WORKING PRINCIPLE :

Servo Motors: Each joint of the robotic arm is actuated by a servo motor. Servo motors are commonly used due to their precise control over angular position

1. Base Rotation (Degree of Freedom 1): The first servo motor controls the rotation of the robotic arm's base. By rotating the base, the arm can orient itself in different directions around a fixed point.
2. Shoulder Movement (Degree of Freedom 2): The second servo motor controls the movement of the arm's shoulder joint. This allows the arm to move up and down or tilt from side to side.
3. Elbow Movement (Degree of Freedom 3): The third servo motor controls the elbow joint of the arm. This joint enables the arm to extend or retract, adjusting its reach.
4. End Effector Rotation or Gripper (Degree of Freedom 4): The fourth servo motor controls either the rotation of the end effector (the tool or attachment at the end of the arm) or the opening and closing of a gripper. This allows the arm to manipulate objects or perform specific tasks.

PWM Control: Pulse Width Modulation (PWM) is a method used to control the position of servo motors. By varying the width of the pulse, you can set the desired angle of rotation for each servo motor. Arduino: The Arduino microcontroller is used to generate PWM signals to control the servo motors. It reads input signals, processes them, and outputs corresponding PWM signals to control the servo motors.

KINEMATICS:

The kinematics of the robotic arm determine how each joint's rotation affects the position and orientation of the end effector (the part of the arm that performs tasks). This involves understanding the relationship between joint angles and the position of the end effector in 3D space. Programming: You need to write a program for the Arduino that interprets desired end effector positions and calculates the required joint angles to achieve those positions. This program will then convert these angles into PWM signals for each servo motor.

CALCULATIONS :

Length of gripper = 6.9cm (DL)

Length of wrist to elbow = 7.7cm (CD)

Length of shoulder to elbow = 8cm (BC)

Length of Base to Shoulder = 4.5cm (AB)

Weight of Motor = 9gm

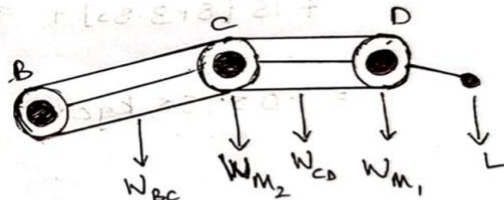
Weight of AB = 17gm

Weight of BC = 25gm

Weight of CD = 15gm

Weight of gripper = 12gm.

Load = 12 + 3 = 15gm.

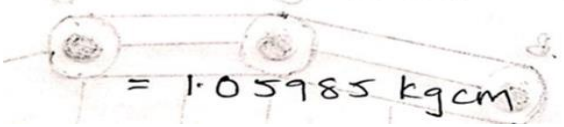


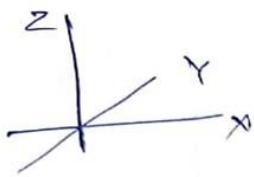
Calculation of Torque

$$\begin{aligned}
 T_D &= (L_A \times L) + \left[\frac{1}{2} \times L_A \times W_A \right] \\
 &= (6.9)(15) + \frac{1}{2}(6.9)(12) \\
 &= 144.9 \text{ gmc} \\
 &= 0.1449 \text{ kgcm}
 \end{aligned}$$

$$\begin{aligned}
 T_C &= L \times [L_{CD} \times L_A] + W_A [L_{CD} \times (L_A/2)] + W_M \times L_{CD} \\
 &\quad + W_{CD} \times (L_{CD}/2) \\
 &= (15)[7.7 + 6.9] + 12[7.7 + 3.45] + 9 \times 7.7 + 15 \times \left(\frac{7.7}{2}\right) \\
 &= 0.47985 \text{ kgcm}
 \end{aligned}$$

$$\begin{aligned}
 T_B &= L \times (L_{BC} + L_{CD} + L_A) + W_A \times (L_{BC} + L_{CD} + (L_A/2)) + \\
 &\quad W_{M_1} \times (L_{BC} + L_{CD}) + W_{CD} \times [L_{BC} + (L_{CD}/2)] + \\
 &\quad W_{M_2} \times L_{BC} + W_{BC} \times (L_{BC}/2) \\
 &= (15)(8 + 7.7 + 6.9) + 12[8 + 7.7 + 3.45] + 9[8 + 7.7] \\
 &\quad + 15[8 + 3.85] + 9 \times 8 + 25\left(\frac{8}{2}\right) \\
 &= 1.05985 \text{ kgcm}
 \end{aligned}$$





$$D = \sqrt{(x_{EE})^2 + (z_{EE})^2}$$

$$\tan \theta_1 = \frac{z_{EE}}{x_{EE}}$$

$$\Rightarrow \theta_1 = \tan^{-1} \left(\frac{z_{EE}}{x_{EE}} \right)$$

$$y_{EE} \geq L_1$$

$$d = D - L_4$$

$$Y_{\text{offset}} = y_{EE} - L_1$$

$$R = \sqrt{(d)^2 + (Y_{\text{offset}})^2}$$

$$\frac{d}{R} = \cos(\alpha_1) \Rightarrow \alpha_1 = \cos^{-1} \left(\frac{d}{R} \right)$$

$$(L_3)^2 = (L_2)^2 + (R)^2 - 2 L_2 \times R \times \cos(\alpha_2)$$

$$\alpha_2 = \cos^{-1} \left(\frac{L_2^2 + R^2 - L_3^2}{2 \times L_2 \times R} \right)$$

$$\theta_2 = \alpha_1 + \alpha_2$$

$$R^2 = (L_2)^2 + (L_3)^2 - 2 L_2 \times L_3 \times \cos(\theta_2)$$

$$\theta_3 = \cos^{-1} \left(\frac{L_2^2 + L_3^2 - R^2}{2 L_2 \times L_3} \right)$$

$$\boxed{\theta_4 = 180^\circ - \left\{ [180^\circ - (\alpha_2 + \theta_3)] - \alpha_1 \right\}}$$

$$y_{EE} \leq L_1$$

$$\theta_2 = \alpha_2 - \alpha_1$$

$$\boxed{\theta_4 = 180^\circ - \left\{ [180^\circ - (\alpha_2 + \theta_3)] + \alpha_1 \right\}}$$

Considering the ~~pos~~ coordinates of drop positions be $(-5, 0, 10)$

$$\Rightarrow D = \sqrt{(-5)^2 + (10)^2} = 11.1803 \text{ cm}$$

$$\theta_1 = \tan^{-1}\left(\frac{10}{-5}\right) = 116.565^\circ$$

$$\rightarrow y_{EE} \geq L_1$$

$$d = 11.1803 - 6.9 = 4.2803 \text{ cm}$$

$$Y_{\text{offset}} = 10 - 4.5 = 5.5 \text{ cm} \quad [\text{Let } y_{EE} = 10 \text{ cm}]$$

$$R = \sqrt{(4.2803)^2 + (5.5)^2} = 6.961 \text{ cm}$$

$$\alpha_1 = \cos^{-1}\left(\frac{4.2803}{6.961}\right) = 52.055^\circ$$

$$\alpha_2 = \cos^{-1}\left(\frac{(8)^2 + (6.961)^2 - (7.7)^2}{2 \times 8 \times 6.961}\right) = 61.495^\circ$$

$$\theta_2 = \alpha_1 + \alpha_2 = 113.550^\circ$$

$$\theta_3 = \cos^{-1}\left(\frac{(8)^2 + (7.7)^2 - (6.961)^2}{2 \times (8) \times (7.7)}\right) = 52.596^\circ$$

$$\theta_4 = 180^\circ - \{[180^\circ - (61.495^\circ + 52.596^\circ)] - 52.055^\circ\}$$

$$\boxed{\theta_4 = 166.146^\circ}$$

DEGREE OF FREEDOM :

$$\text{Degree of freedom} = 3 (n - 1) - 2j - h$$

Where, n = number of links j =

number of lower pairs

h = number of higher pairs

In this case of robotic arm,

$$n = 5$$

$$j = 4$$

$$h = 0$$

$$\text{Therefore, degree of freedom} = 3 (5 - 1) - 2 (4) - 0$$

$$= 3 (4) - 8$$

$$= 12 - 8$$

$$= 4$$

Hence the degree of freedom of the specified robotic arm is 4.

FUTURE ADVANCEMENTS :

The use of robotic arms, systems and devices for surgery began in the early 80s, yet their ability to assist surgeons with difficult tasks with greater accuracy is becoming an increasingly accepted practice in operating rooms across the globe.

By extending human surgeons ability to plan and carry out surgical interventions more accurately and less invasively, surgical robotic systems can address a vital international need to greatly reduce costs, improve clinical outcomes, and improve the efficiency of health care delivery

The impact of robotic systems and devices on surgery in the next 25 years promises to be as comparable to that of manufacturing robots on industrial production over the past 25 years

ADVANTAGES :

Efficiency: They can work continuously without fatigue, leading to increased productivity and reduced cycle times.

Safety: By handling hazardous materials or operating in dangerous environments, robotic arms enhance workplace safety by minimizing the risk to human operators

Flexibility: Robotic arms can be reprogrammed or reconfigured to perform different tasks, making them adaptable to changing production needs.

Cost-effectiveness: While initial investment costs may be high, robotic arms often provide long-term cost savings through improved efficiency, reduced waste, and lower labor costs.

Increased Output: Robotic arms can work faster and for longer durations than human workers, leading to higher output levels and greater throughput

DISADVANTAGES :

Limited Reach: A four-DOF robotic arm may have constraints in reaching certain positions or objects within its workspace due to its restricted range of motion along different axes.

Limited Manipulation: With fewer degrees of freedom, the robotic arm may struggle to manipulate objects in complex ways, such as grasping items from various angles or performing intricate assembly tasks.

Difficulty in Avoiding Obstacles: With fewer joints and degrees of freedom, it may be more challenging for the robotic arm to navigate around obstacles or obstacles within its workspace, potentially leading to collisions or inefficiencies.

Reduced Flexibility: The arm's ability to adapt to different tasks or environments may be limited compared to robotic arms with higher degrees of freedom, which can hinder its versatility in certain applications

FURTHER MODIFICATIONS :

robotic arm can also be controlled using a Bluetooth controller. This involves connecting the Bluetooth controller to the robotic arm's control system, allowing users to wirelessly send commands to control its movements and actions. This method provides more flexibility and convenience in controlling the robotic arm, especially in applications where direct physical interaction with the arm is not feasible or practical.

We can also control a robotic arm using a potentiometer controller by connecting potentiometers to the joints of the arm. As the potentiometers are turned, they generate varying voltage signals corresponding to the position of each joint. These voltage signals are then processed by a microcontroller or a control system to adjust the position and movement of the robotic arm accordingly. This method provides a manual control interface, allowing users to directly manipulate the arm's movements by adjusting the potentiometers

CONCLUSION :

The robotic arm boasts versatility and precision, enabling seamless movements in multiple directions. It efficiently harnesses PWM signals and an Arduino microcontroller for nuanced control, ensuring accuracy in its operations. With predefined task execution capabilities, it enhances automation by streamlining processes and minimising manual intervention, thereby optimising productivity. Representing a sophisticated solution for modern automation needs, the robotic arm exemplifies the fusion of advanced technology and practical efficiency, offering a reliable and adaptable tool for a variety of industries and applications.